Upify: Predicting Loan Defaults Using Everyday Spending Behavior

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Abstract—Every year banks lose an enormous sum of money due to borrowers refusing to repay their loans. Banks use old data (e.g., past credit scores; repayment history) to identify those borrowers who might be risky. Often, when a bank makes the decision about whether to lend a borrower money, this data is months old. We have developed a solution to the problem of identifying risky borrowers called Upify. The Upify solution allows a bank to evaluate the current spending habits of its potential borrowers as recorded by UPI (India's digital payments network), in conjunction with loan and geographical location data. Based on this information, Upify can predict whether or not a potential borrower will cease making loan repayments. More importantly, Upify attempts to address a greater question: did the borrower truly run out of money or did the borrower simply decide not to pay their loan payments although they're able to? We refer to the two types of borrowers as genuine.

defaulters and *intentional defaulters*. We tested the system on 45,000 customer records and reached 88.7% accuracy about 14–15 percentage points better than systems using only old financial data. It correctly identified 91.3% of genuine defaulters and 87.6% of intentional defaulters.

Index Terms—Loan Default Prediction; UPI Transaction Analytics; Behavioral Finance; Random Forest; Credit Risk; Intentional vs. Genuine Defaulters; Feature Engineering; India; SMOTE

I. INTRODUCTION

II. BACKGROUND AND RELATED WORK

A. The History of Credit Scoring

Credit scoring has existed for decades. The first credit scoring models were based on financial ratios (i.e., how much debt a corporation has compared to how much net income it generates) to predict whether or not a corporation would declare bankruptcy. Banks in previous decades employed logistic regression to develop credit scores for consumer lending because of its simplicity; however, the limitation of this type of methodology is that it assumes the applicant's past performance is indicative of future behavior, which fails when an applicant's circumstances change very quickly.

With machine learning, the accuracy of the predictions will increase. Studies have shown that tree-based models (e.g., Random Forest) outperform other models by 5-10% over traditional credit scoring methods [3].

B. Advanced AI Models

Recently, LSTM creations have begun to create better predictions by analyzing mobile transaction data [5]. However, these models require very large sample sizes and a vast amount of computational power, which most mid-size lenders in India do not possess.

C. Alternative Data and Behavioral Patterns

Research shows that individuals who use digital methods to conduct financial transactions, such as mobile payments, demonstrate behavioral attributes similar to those specified by traditional credit scoring methods, and in some cases, may produce even more accurate results than traditional credit methods [11, 12]. Our main area of research is focused on analyzing UPI transaction data because UPI transaction data is more abundant in India compared to other mobile financial transaction data.

D. Gaps in Existing Literature

The current body of literature offers limited insight into the predictive nature of UPI via financial behavior. The overwhelming majority of researchers that have been published have only used traditional methods (income, credit history, employment, loan amount) of predicting credit risk. As a result, there are no known studies reflecting the use of UPI transactions in real-time for the purpose of predicting the credit risk of consumers. separated intentional defaulters from genuine ones. And most models are hard to explain to borrowers or regulators. Upify addresses all three gaps.

III. SYSTEM DESIGN AND METHODOLOGY

Data collection, cleaning, feature building, prediction, and dashboarding comprise the five stages of Upify. At their core, these stages have an established purpose and can operate independently of each other, allowing for streamlined updates.

A. Data Collection

The Upify platform can collect four data types including (1) Loan Records (Loan Amount (e.g., $30,000), Payment Schedule (e.g., Sch. A) and Payment History (e.g., 5 Delinquencies); (2) 12 months of UPI Transaction Logs (Transaction Amount, Merchant Type, Merchant GPS Location); (3) Basic identifying information (Age, Occupation, Income Bracket, Home District); (4) Unique Customer Identification Code to protect customer privacy. Each record in the database is tied together using a unique (but scrambled) identifier. Timestamp differences between sources will be reconciled automatically.

B. Preprocessing

Missing numeric data are filled with the average value of all the numeric data available from the other three sources to ensure accuracy of analysis. All numeric data will be scaled within a standard range to allow equal representation of the two extreme ends of the numeric data range (e.g., loan amounts may be significantly larger than transaction amounts). In order to create a supportive document format to use in subsequent model development, we will convert all text-based category data to a format that can be utilized by the model. Any flagged or irregular transactions will be maintained because they may represent crucial signals for the model.

C. Feature Engineering

The five (5) behavioral features developed constitute the heart of Upify. Each of them isolates a specific aspect of a subject's past and/or present financial behavior that was not available in traditional credit reporting sources.

TABLE I. Behavioral Features and Their Predictive Rationale

Feature	Definition	Why It Matters
Spending Ratio	Luxury spend ÷ essential spend	Ratio > 1.5 while missing payments suggests choosing not to pay
Transaction Frequency	UPI payments per 30-day window	Sharp drop often comes 60–90 days before first missed EMI
Monthly Spending Pattern	Month-on-month variance by category	Big swings signal unstable income or erratic habits
Income-to-Loan Ratio	Monthly income ÷ monthly EMI	Below 1.3 means the borrower is stretched too thin
Location Spread	Std. dev. of GPS coords across transactions	Wide spread + luxury spend + missed EMIs may indicate asset concealment

D. Machine Learning Models

We used 2 models for training purposes. The first was the Logistic Regression which acted as a transparent baseline. The second model was a Random Forest with 200 trees, a max of 12 depth levels and a 5-fold cross validation to tune the parameters.

E. Genuine vs. Intentional Classification

All of the predicted defaulters had a second check performed via rules - which left little to the imagination for the loan officers or borrowers to determine.

* Genuine Defaulter: A majority of their monthly expenditure is on necessities & they have little to no other options available to them - therefore making their payment obligation exceeds their monthly expenses & they cannot pay.

* Intentional Defaulter: They typically have excessive luxury spending (i.e. > 30%) of their total monthly expenditure (i.e. loan) & based on their income to amount of loan are able to pay their loan; however, they have intentionally selected to use the excess funds elsewhere.

IV. EXPERIMENTS AND RESULTS

A. Dataset

The dataset includes 45k anonymized loans from a cooperative bank in 3 districts of Uttar Pradesh for each borrower for a period of 24 months. 31% of all borrowers (13,950) defaulted. 58% of the defaulted borrowers were legitimate defaulters and 42% were intentional defaulters based on the agreed upon terms of the loan and verified through legal case outcomes and the review of a company's credit officer. The dataset was divided into 80 % for training and 20 % for testing, in a stratified basis.

B. Primary Classification Results

TABLE II. Model Performance (highlighted row = Upify system)

Model	Acc.	Prec.	Recall	F1	AUC
LR (static data only)	74.2%	71.8%	68.5%	0.701	0.789
LR (all features)	79.6%	77.3%	75.1%	0.762	0.841
RF (static data only)	80.4%	78.9%	76.2%	0.775	0.858
RF + Upify features	88.7%	87.4%	86.1%	0.867	0.934

Just adding UPI features to the basic model pushed accuracy up 5.4 points. The full Upify system reached **88.7% accuracy and AUC-ROC of 0.934**.

C. Confusion Matrix

TABLE III. Confusion Matrix (test set, 9,910 records)

	Predicted: Default	Predicted: No Default
Actual: Default	TP = 2,451	FN = 396
Actual: No Default	FP = 311	TN = 6,752

With 396 missed defaulters against 2,451 correctly caught, recall improved from 68.5% (static baseline) to **86.1%** — a significant gain in identifying risky borrowers.

D. Feature Importance

TABLE IV. Random Forest Feature Importance Scores

Feature	Importance
Spending Ratio	27.3%
Transaction Frequency	19.8%
Income-to-Loan Ratio	16.2%
Monthly Spending Pattern	13.1%
Location Spread	9.4%
Credit Score (static)	7.6%
Loan-to-Income Ratio (static)	6.6%

Behavioral features account for **85.8%** of total model importance. Static credit features contribute only 14.2%.

E. Second-Stage Results

91.3% of accurate default cases were valid; 87.6% were also accurately identified as intentional defaults. The second stage of the operational process achieved a total accuracy of 89.7%. Only 8.3% of the submitted terms required a human to review them – an operationally efficient outcome.

V. SYSTEM MODULES

A. Data Collection Module

Data is pulled from the Banking system, the UPI log API's demographic and GPS data every night (runs nightly and on occasion) and any records that are not a perfect match across source systems are flagged (not guessed) and will show up on a daily data quality report.

B. Feature Engineering Module

Calculates five behavioral signals every month. The typical time frame is 12 months, but institutions with a shorter history can calculate using 3 or 6 months.

Loads the trained Random Forest and produces a risk score (0–1) per customer, plus SHAP-based explanations for the top three factors driving that score — visible on the dashboard.

C. Behavioral Classification Module

Applies the genuine vs. intentional rules to predicted defaulters. Every output includes the specific feature values and threshold comparisons that produced it — fully transparent and auditable.

D. Dashboard Module

A web interface where loan officers browse risk scores, behavioral labels, and 12-month spending trend charts. They can drill down to individual transactions. A portfolio view shows the distribution of risk across the entire active loan book.

VI. COMPARATIVE ANALYSIS

TABLE V. Comparison with Representative Prior Work

Study	Data	Model	Acc.	Intent ?
Altman [1]	Financial ratios	Z-Score	~72%	No
Lessmann et al. [3]	Credit bureau data	Random Forest	~79%	No
Xia et al. [4]	Credit bureau data	XGBoost	~82%	No
Wang et al. [5]	Sequential transactions	LSTM	~84%	No
Upify (Ours)	Loan + UPI + GPS + Demo	Random Forest	88.7%	Yes

There is a significant gap between the columns regarding how much effort there is to differentiate between a believer and a reason-oriented human. In terms of accuracy alone, Upify is outperforming LSTM deep learning on raw accuracy, despite using a simpler model than LSTM: Our behavioral features identify similar behaviour but in a more literal, less computationally costly way.

VII. DISCUSSION

A. What the Results Mean in Practice

The intuition behind the Spending Ratio being by far the most predictive feature makes good common sense: If someone has higher relative luxury spending versus basic spending than someone else, then it's hard to say that they are suffering from Financial Hardship if they don't pay their loan. Credit models are not taking advantage of the very large predictive signal we have identified in them.

The distinction between genuine versus intentional defaulting has very significant practical implications for financial institutions. For someone who is a genuine defaulter, instituting aggressive legal actions is usually not the best way to resolve the situation; instead, loan modifications or a pause in payments will usually provide a much better outcome.

The Location Spread metric appeared consistently in high-value intentional default cases. Borrowers transacting across widely dispersed locations — particularly at luxury venues — while missing EMIs represent a profile that spending features alone would not catch.

B. Limitations

- Data from a single institution in one region. Results maynot generalize to commercial banks or rural microfinance lenders without further validation.
- The rule-based second stage is deliberately simple. The12% of borderline cases sent for human review represent genuinely hard calls fixed rules cannot resolve.
- Depends on UPI adoption. Cash-heavy borrowersproduce thin or no UPI history; the system falls back to static features for them.
- Location data raises consent and privacy questionsaddressed in Section VIII.

VIII. ETHICAL CONSIDERATIONS

A. Fairness and Bias

Data about spending may also unintentionally highlight socioeconomic conditions rather than purchasing behaviour. A borrower living in an area of low-income may have a high Spending Ratio simply because the purchases are being classified as Non-Essential; whereas in reality, they are basic goods being purchased from stores. We conducted a fairness audit based on income decile, age, and gender. There was no statistically significant differential in False Negative rates ($P = 0.17$). However, a broader, multi-institution audit may be needed.

B. Privacy and Consent

Generally speaking, the use of GPS data in credit decision-making is relatively new. In our case, it was established that the institution had a valid, explicit consent agreement for transaction monitoring. As such, we believe production deployment will require obtaining explicit informed consent for the use of GPS-based analysis, and borrowers being able to review their individual behavioural profiles, and a mechanism to waive is implemented - items which will most likely be a requirement of the implementation of the DPDPA 2023 in India.

C. Interpretability and Contestability

A credit decision that cannot be understood or explained cannot be contested. Upify has a second stage of decision-making based on 'rules', which was deemed preferable to using a more precise 'learned' model because it provides decisions with plain language explanations, e.g., "You have a 2.3 Luxury Spending Ratio but only a 1.8 Income-to-Loan Ratio, indicating that you have the ability to repay a loan but are not likely to repay." This kind of plaintext of credit-related explanations can facilitate further engagement with the credit evaluators as well as the borrowers.

IX. FUTURE DIRECTIONS

You cannot challenge credit resolution if it was not able to be understood or explained.

Rather than using a 'learned' decision method (e.g., using models developed from historical data), we have a second stage of decision-making use of 'rules.'

Because this allows credit decisions to be made using plain language (e.g., the consumer has a 2.3 luxury spending ratio as compared to the consumer's income-to-loan ratio of 1.8.

Therefore, the consumer could repay the loan; however, there is a very low probability that the consumer will repay the loan) a plaintext explanation for credit-related resolution could further incentivize continuing the credit evaluation process for both the credit evaluator and the credit consumer.

X. CONCLUSION

The main point is fairly straightforward: the way a person spends their money at present indicates much more accurately whether they will pay back a loan than if you look back three years to see what the person's previous spending looked like. With the introduction of Unified Payments Interface (UPI), this real-time data was made available across India at an unprecedented scale, yet has not been used widely to predict the credit risk associated with customers. Upify is an initiative aimed at changing that practice.

By constructing five behavioral characteristics from UPI User Data and combining these behavioral features with standard loan variables to create a Random Forest model, we achieved an 88.7% accuracy rate (on 45,000 records), approximately 14-15% higher than with static-feature benchmarks. Additionally, our second-tier classification of a customer's intent to repay their loan allows us to provide a principled (auditable) basis for determining the reason for potential defaulting behavior.

Neither of these achievements is technically difficult. Random forests and rule-based classifiers are common data-mining techniques used extensively in the industry. However, what is novel is the method applied to conduct the analyses and the information contained in the data set. The resulting analysis provides compelling evidence that analysis of behavioral transactions should be a core component of evaluating credit risk in India rather than an auxiliary component

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